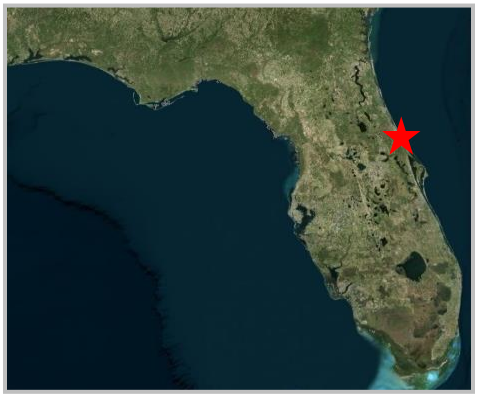
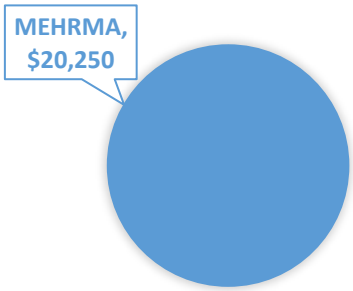
City Park Living Shorelines, City of Edgewater, Volusia County, FL

FY 16/17



Project Partners and Funding



Project Description

The City Parks Living Shorelines-City of Edgewater project is a result of a largescale Living Shorelines initiative which has been conducted throughout the Indian River Lagoon (IRL) over the past decade. FWC has partnered with regional entities (municipalities, Marine Discovery Center-MDC, University of Central Florida-UCF) to conduct Living Shorelines projects, sometimes using MEHRMA funding. Other regional entities (e.g., Florida Oceanographic Society, Brevard Zoo) have also played a role in these types of enhancements. As a result, the IRL restoration community has perfected techniques for improving IRL shorelines in terms of using natural products to secure erosion.



Figure 1 Typical Indian River Lagoon coquina rip-rap with native plants intermixed.

The project is a partnership between FWC and the City of Edgewater. During 2016/17, the partners identified two city parks along Mosquito Lagoon in the IRL experiencing erosion. General conceptual plans were developed to secure the erosion through Living Shoreline designs. Living Shorelines emphasize the use of natural products such as native stone and plants as opposed to traditional hardening techniques such as seawalls. The funding provided by MEHRMA for the City Parks project was used to purchase native coquina rock which will be used along the park waterfront as breakwaters to reduce wave energy and promote sediment accretion and plant survivorship. The coquina and native plants provide habitat for aquatic organisms.

Benefits

- Provide a natural breakwater to reduce wave energy along Indian River Lagoon

- Increased community resiliency to the effects of sea level rise
- Promote sediment accretion and protect native plants
- Provide habitat for aquatic organisms in a Living Shoreline

FWC Strategic Plan Elements Supported

Theme 1, Goal 2: Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategy 3. Manage habitats to sustain healthy and diverse fish and wildlife populations

Acreage restored by habitat type (footprint)

This project entails the planning for Living Shorelines along city parks in Indian River Lagoon:

- Hundreds of linear feet of Living Shorelines (estimated over 0.12 acres)
- Native coquina stone and plants provide habitat for aquatic organisms

Acreage affected by restoration activities: 5 acres (public lands protected)



Figure 2: Example of a rock breakwater causing sediment accretion.



Figure 3: Native plants securing an IRL shoreline and providing habitat.